

Flowchart Sistem — Tabungan Haji API

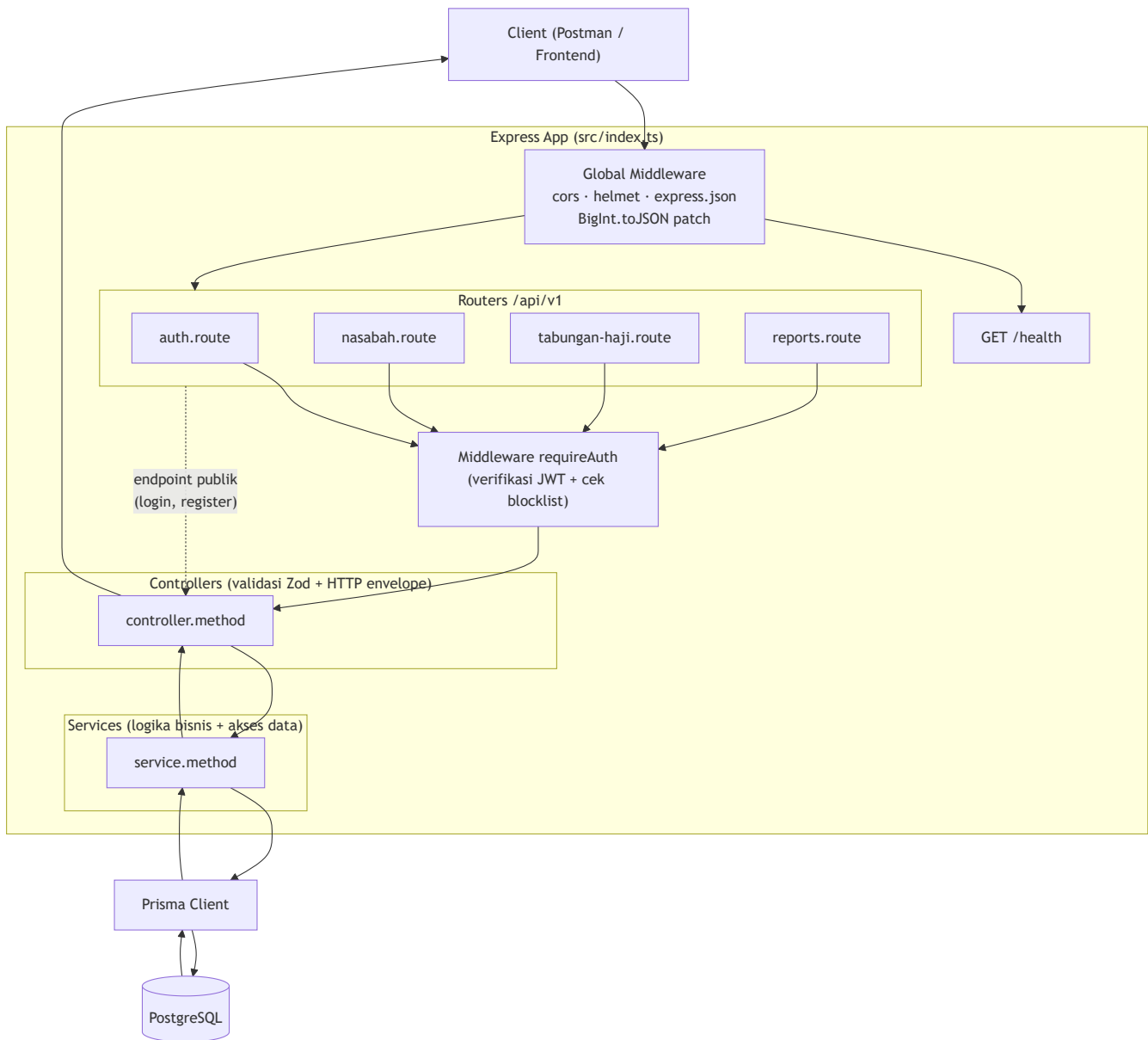
Dokumen ini berisi diagram alur sistem dalam format **Mermaid**.

Untuk melihat render-nya: buka di GitHub, atau di VSCode pakai ekstensi **Markdown Preview Mermaid Support**.

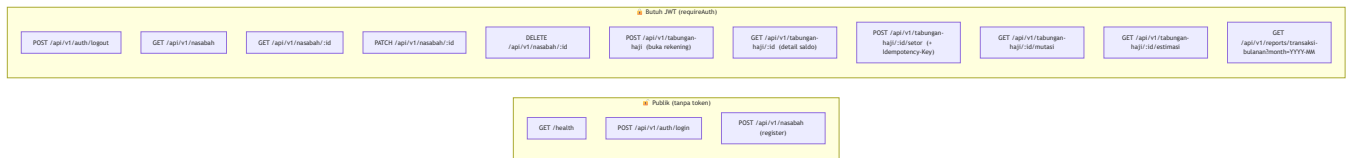
Stack: **Express 5 + TypeScript + Prisma 6 + PostgreSQL + Zod + JWT**.

1. Arsitektur Berlapis (Layered Architecture)

Tiap request mengalir lewat lapisan yang sama: Route → (Auth) → Controller (validasi) → Service (logika + DB) → Prisma → PostgreSQL.

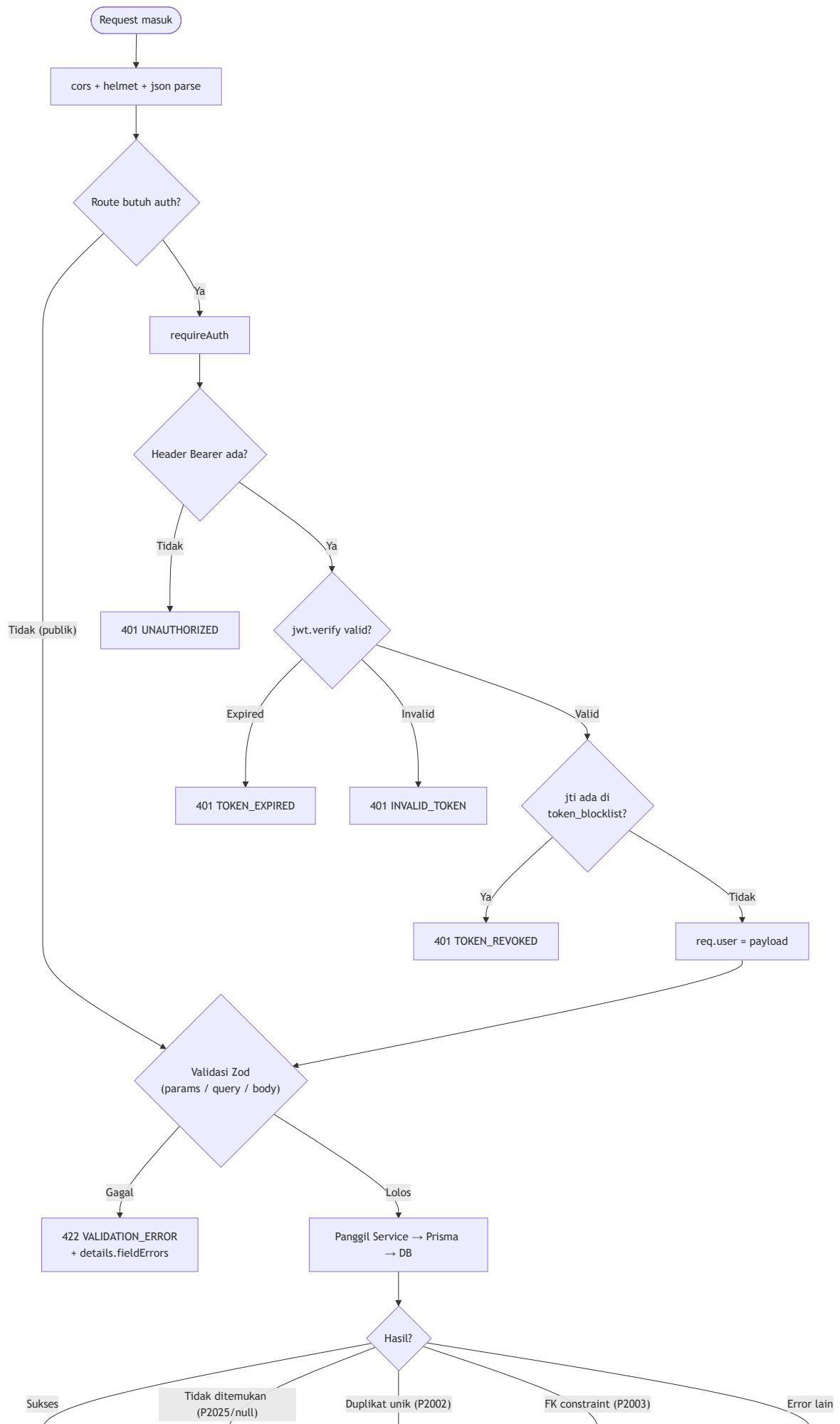


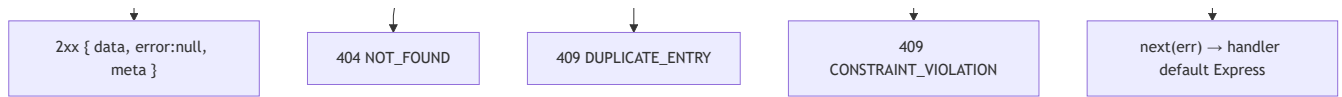
2. Peta Endpoint & Proteksi



3. Lifecycle Request Umum (jalur ter-proteksi)

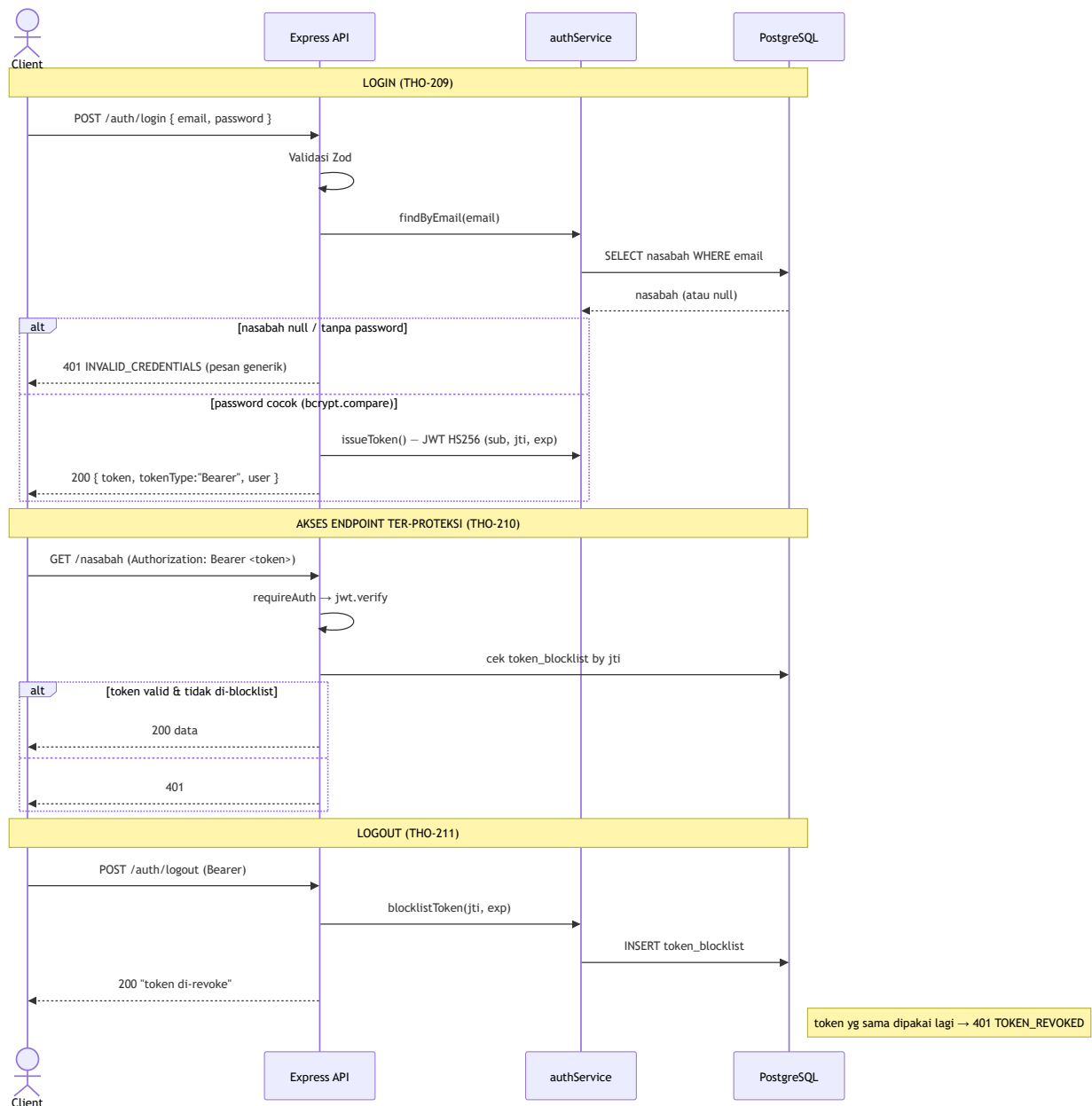
Pola yang dipakai semua controller: validasi dulu, baru sentuh DB, lalu map error → HTTP.





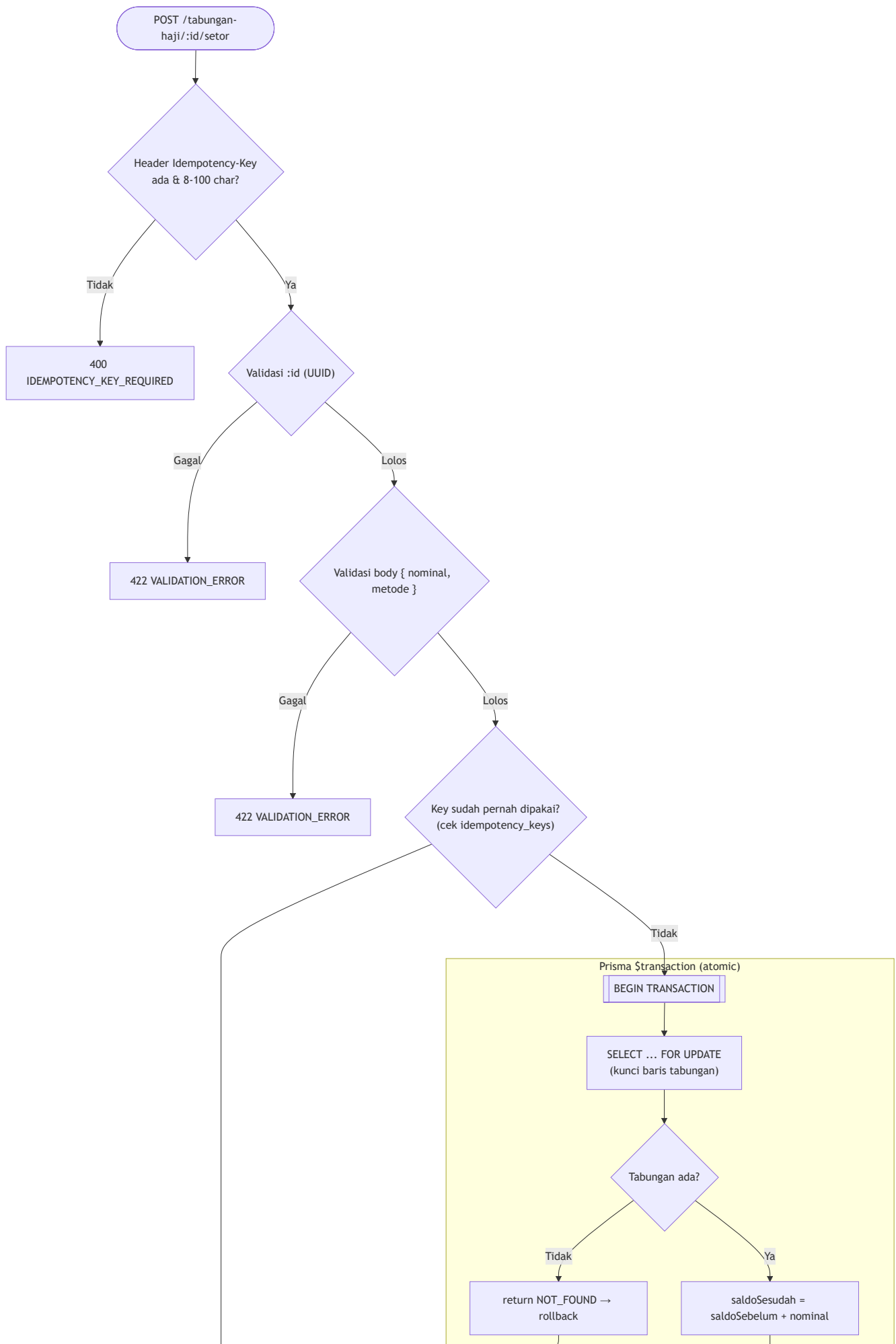
Semua response sukses & error memakai envelope konsisten: { data, error, meta:{ timestamp } } .

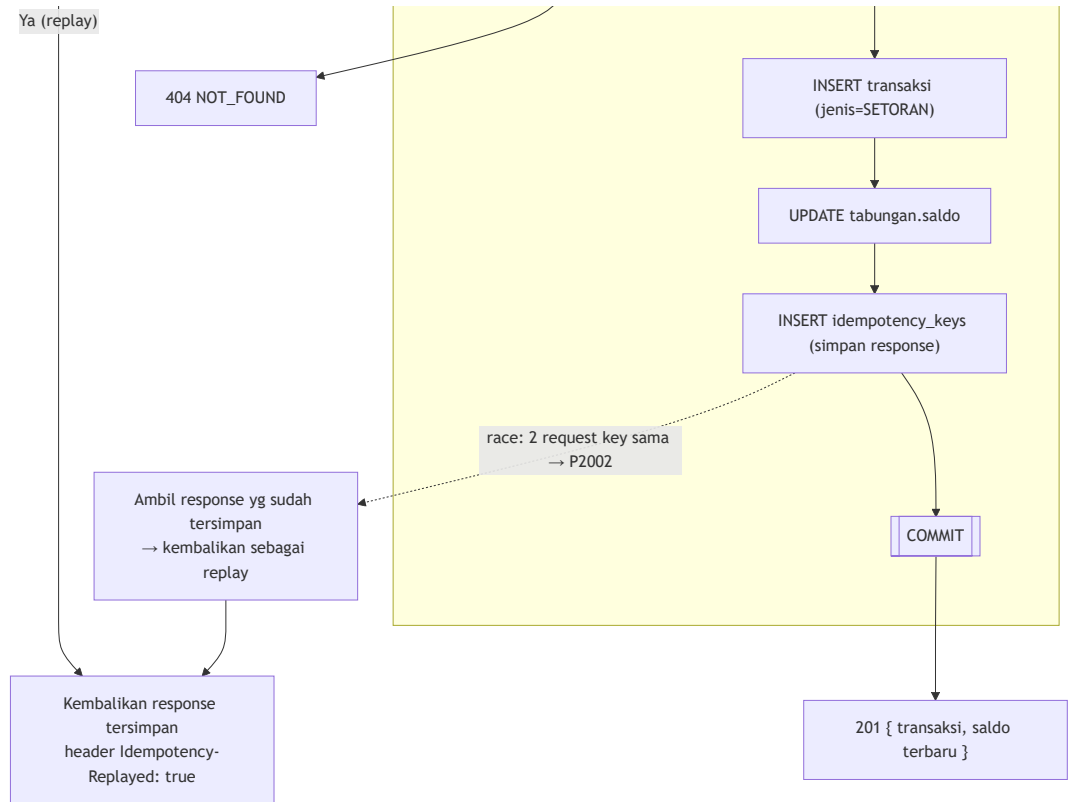
4. Flow Autentikasi (Login → Pakai Token → Logout)



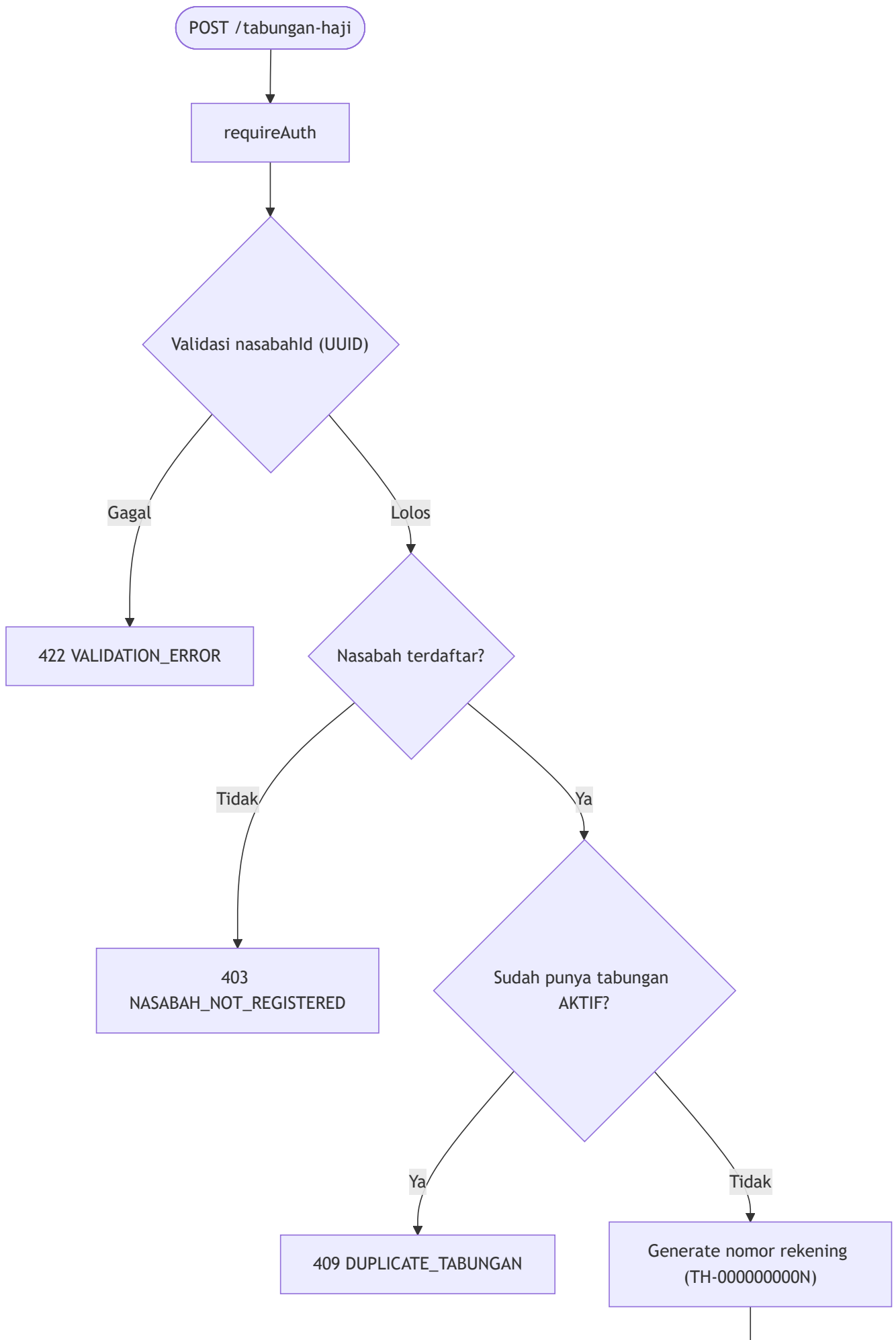
5. Flow Setor Saldo (THO-206) — Idempotency + DB Transaction

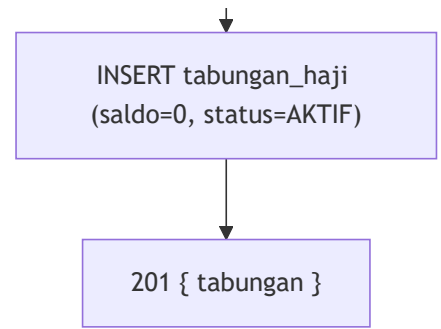
Endpoint paling kompleks. Pakai **Idempotency-Key** (anti dobel-setor) + **transaksi DB** dengan **row lock** `FOR UPDATE` (anti race / double-spend).





6. Flow Buka Rekening (THO-205)



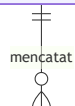


7. Model Data (ERD)

NASABAH			
uuid	id	PK	
string	nik	UK	16 digit
string	nama		
string	email	UK	
string	nomor_hp		
string	password		bcrypt hash, nullable
datetime	created_at		
datetime	updated_at		



TABUNGAN_HAJI			
uuid	id	PK	
uuid	nasabah_id	FK	
string	nomor_rekening	UK	
bigint	saldo		default 0
string	status		AKTIF / ...
datetime	dibuka_at		



TRANSAKSI			
uuid	id	PK	
uuid	tabungan_id	FK	
string	jenis		SETORAN / ...
bigint	nominal		
bigint	saldo_sebelum		
bigint	saldo_sesudah		
string	referensi	UK	
string	metode		nullable
datetime	waktu		

IDEMPOTENCY_KEYS		
string	key	PK
string	endpoint	
int	status_code	
json	response	
datetime	created_at	

TOKEN_BLOCKLIST			
string	jti	PK	
uuid	nasabah_id		nullable
datetime	expires_at		
datetime	created_at		

IDEMPOTENCY_KEYS & TOKEN_BLOCKLIST berdiri sendiri (tidak ada FK formal ke nasabah) — dipakai sebagai mekanisme idempotensi setoran dan revoke token JWT.